

Dev Jariwala

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EDUCATION

University of Michigan

B.S.E. in Computer Science

Ann Arbor, MI

May 2028

Relevant Coursework: Data Structures and Algorithms, Foundations of Computer Science, Discrete Mathematics

EXPERIENCE

Hyundai America Technical Center (HATCI) via KPIT Technologies

Novi, MI

Software Engineer Intern, Client Assignment

July 2026 – Aug. 2026

- Developed **20+ Robot Framework automated test cases** using **Python** and **Linux**, expanding regression coverage across **5 Hyundai ccNC infotainment modules** and SiriusXM 360L services.
- Validated embedded infotainment release builds, investigated software defects, and documented repeatable testing procedures across **5 Hyundai ccNC modules** in collaboration with a **5-person KPIT team**.
- Performed manual in-vehicle testing of Hyundai ccNC infotainment features while supporting HATCI's **organization of 100+ engineers**, supplementing automated regression testing and verifying system behavior.

KPIT Technologies

Novi, MI

Software Engineer Intern

June 2026 – July 2026

- Built an internal authentication and user-management platform using **Angular**, **Flask**, and **MySQL** with secure **JWT authentication**.
- Implemented **6 REST APIs** integrated across **10+ frontend pages**, supporting authentication, user management, and role-based access control.

Necessities

Ann Arbor, MI

Software Engineer Intern

Sep. 2025 – Apr. 2026

- Built **50+ React components** and PostgreSQL APIs supporting **100+ active users**, enabling marketplace search, filtering, messaging, and asynchronous product updates.
- Improved application performance using lazy loading, virtualized rendering, and PostgreSQL query optimization across a marketplace containing **1,000+ product listings**.
- Delivered biweekly production releases within a **5-person engineering team** using Git, pull requests, code reviews, and automated CI pipelines.

PROJECTS

MiniRedis | C++20, TCP/IP, Multithreading, CMake

- Built a Redis-style in-memory key-value database in **C++20** implementing **6 core Redis commands** (SET, GET, DEL, EXISTS, EXPIRE, TTL) over concurrent TCP client connections.
- Implemented thread-safe storage using reader-writer locks, TTL expiration, LRU eviction, a fixed-size worker pool, and append-only persistence for crash recovery.
- Developed unit and integration tests covering parsing, concurrency, persistence, and expiration; benchmarked **25,000 requests** at **52,000+ operations per second**.

Cricket Scoring & Analytics Platform | React, TypeScript, Supabase, PostgreSQL, Google Cloud

- Built a real-time cricket platform supporting **3 synchronized workflows**: match scoring, player statistics, and live game-state updates through WebSockets.
- Implemented a scoring engine supporting **15+ tournament rules**, including dismissals, extras, strike rotation, player statistics, and dynamic second-innings target calculations.
- Protected application data through **3 security layers**: OAuth 2.0, PostgreSQL row-level security, and middleware enforcing role-based authorization.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL

Frameworks/Libraries: React, Angular, Flask, Robot Framework, Flutter

Databases: PostgreSQL, MySQL, Firestore, Supabase

Cloud/Developer Tools: Google Cloud Platform (GCP), Docker, Git, Linux, Vercel, Jumpstarter, CMake, Ollama